Roll Number: _____

Thapar Institute of Engineering and Technology,
Patiala Department of Computer Science and Engineering

B E- COE, CSE (Summer Sem)
MST Course Code: UCS701 — Course Name: Theory of Computation
July 1st, 2026, 3:30 PM — Time: 2 Hours, M. Marks: 30
Name Of Faculty: Dr. Nidhi Chakravarty

Note: Attempt all questions with proper justification. Assume missing data, if any, suitably.

S.no	Questions	Marks
Q1	In each case below, find a string of minimum length in$\{a,b\}^*$ not in the language corresponding to the given regular expression. **a)** $b^*(ab)^*a^*$ **b)** $(a^*+b^*)\ (a^*+b^*)\ (a^*+b^*)$ **c)** $a^*(baa^*)^*\ b*$ **d)** $b^*(a+ba)^*b^*$	1*4= 4
Q2	Find a regular expression corresponding to each of the following over {a,b}. **a)** The language of all strings containing exactly two a's. **b)** The language of all strings containing at least two a's. **c)** The language of all strings in which the number of a's is even. **d)** The language of all strings that begin or end with aa. **e)** The language of all strings containing the substring aa.	1*5= 5
Q3	Draw an DFA accepting the indicated language over {a,b}. **a)** The language of all strings in which both the number of a's and the number of b's are even. **b)** The language of all strings that begin with aa or end with aa.	2.5*2 =5
Q4	Given the regular expression $r = (101\|10)^*$ **a)** Convert the given r into NFA using Thompson's construction. **b)** Convert the obtained NFA into DFA. (show all intermediate steps) **c)** Minimize the obtained DFA	2+4+2= 8
Q5	**a)** Differentiate Mealy and Morre machine in terms of tuples. **b)** Design a Moore machine with the input alphabet {0, 1} and output alphabet {Y, N} which produces Y as output if input sequence contains 1010 as a substring otherwise, it produces N as output. **c)** Convert it into an equivalent Mealy machine. (show all intermediate steps)	2+3*2= 8